

CO. ASSESSMENT CASE STUDY:

① → Scenario:

A forensic investigator is reviewing the evidence handling records for a cybercrime investigation involving a seized external hard drive. Examine the evidence log below and identify the point at which the chain of custody may have been compromised. Explain why this issue could affect the admissibility of the digital evidence in court.

Evidence Log:

Date

2026-05-18
09:15 External hard drive (item #102) seized by officer A at the suspect's residence.

2026-05-18
09:25 SHA-256 hash value computed and recorded by officer A. 9f2d....

2026-05-18
09:35 Evidence sealed and transferred to the digital evidence storage room.

2026-05-19
11:40 Evidence storage register shows the seal was opened for routine inspection, but no officer name or authority was recorded.

2026-05-20
8:30 Forensic Analyst B received the hard drive for imaging.

2026-05-20
8:35 SHA 256 hash verified: 9f2d... (match)

2026-05-20
12:10 Forensic image created, original evidence resealed, and returned to the evidence locker.

→ Questions:

1. Identify the point at which the chain of custody broken.

3. What documentation or procedure should have been followed to maintain evidence integrity?
4. Suggest best practices to prevent similar chain of custody violations in future investigations.

Analysis:

1) Chain of custody was broken at 2016-05-19. The evidence was opened for inspection, but no officer name or authorization was recorded.

2A) → Unauthorized access creates doubt about who handled the evidence
→ The defense may claim the evidence was tampered with or altered.

3) → Record officer's name, signature, date & time
→ State the reason for opening the seal.
→ Obtain proper authorization.
→ Record resealing details.

4) Best practices:

→ Allow only authorized personnel to access evidence.
→ Maintain complete chain of custody records.
→ Use tamper-evident seals.
→ Conduct regular audits and verifications.

② Analytical problem 2. missing evidence documentation:

→ During a cyber fraud investigation - seized a suspect's smartphone believed to contain important digital evidence. Review the evidence log below and identify where the chain of custody was compromised. Explain how this could affect the admissibility of the evidence in court.

- 2026-08-12 10:20 Smart phone (item #56) seized by officer A during the investigation.
- 2026-08-12 10:30 SHA-256 hash value computed and recorded by officer R:8als.
- 2026-08-12 10:40 Evidence sealed and transferred to the digital evidence storage room.
- 2026-08-13 9:15 Evidence is issued for forensic examination. Transfer form signed by Custodian.
- 2026-08-13 9:20 Evidence received by forensic Analyst M.
- 2026-08-13 9:30 SHA 256 hash verified. 8als. (match)
- 2026-08-13 12:00 forensic image created and original smart phone returned to evidence locker.

Questions:

1. identify the part of chain of custody broken?
2. Why would the missing document raise concern during legal proceedings?
3. What information should have been recorded to maintain a complete chain of custody?
4. Suggest best practices to avoid such documentation in future.

Answers:

- 1A) 2026-08-13 9:20 - forensic Analyst M received the evidence, but receiver signature was missing.
- 2A) → There is no proof that the evidence was officially received.
→ Accountability is missing, making the evidence questionable.
- 3A) → Receiver's name, Signature, Date, time, purpose of transfer.
→ condition of evidence.
- 4A) → Ensure every transfer is signed.
→ use standardized custody forms.
→ verify documentation before accepting evidence.
→ conduct regular record checks.

③ ⇒ Hash value mismatch During Evidence Verification
→ A forensic team is investigating a cyber intrusion involving a server. A desktop computer belonging to the suspect was seized for forensic analysis.

Evidence Log

2026-09-15 08:30	Desktop computer seized by officer K from the suspect.
2026-09-15 08:40	SHA-256 hash completed and recorded by officer K-Fad
2026-09-15 09:00	Evidence sealed and placed in the secure evidence locker.
2026-09-16 10:15	Evidence released to forensic Analyst P for imaging.
2026-09-16 10:20	SHA-256 hash before imaging CSDA.
2026-09-16 10:25	imaging continued despite the hash mismatch.

Questions:

1. Identify the point where the chain of custody of evidence?
2. Why is the hash mismatch a serious concern during a forensic?
3. Should forensic analyst have continued imaging the device?
4. What corrective action should have been taken immediately.

Answers:

- 1A) 2026-09-16 10:20 - SHA-256 hash did not match the original.
2026-09-16 10:25 - imaging continued despite the mismatch.
- 2A) → indicates possible alteration or corruption of evidence.
→ integrity of the digital evidence cannot be trusted.
- 3A) → No, imaging should stop immediately until case of the mismatch.
- 4A) → Stop forensic examination.
→ Report the mismatch immediately
→ Recalculate the hash.
→ Recalculate possible tampering.
→ Document all findings before proceeding.

Improper Evidence Storage:

→ During a cyber espionage investigation, a suspect's USB flash drive containing confidential company data was seized. Review the evidence log & where the chain of custody was compromised.

Evidence Log:

2026-10-08 14:10 USB flash Drive seized by officer S from suspect.
2026-10-08 14:20 SHA 256 hash computed and recorded by officer.
2026-10-08 14:30 Evidence photographed, labeled and sealed in an evidence locker.
2026-10-08 15:00 Due to the evidence locker being full, the sealed evidence was placed in a secure evidence locker.
2026-10-09 17:00 Evidence transferred to the secure evidence locker.
2026-10-09 19:00 Forensic Analyst Q retrieved the USB drive.
2026-10-09 11:30 SHA-256 hash verified. All good.

Questions:

1. Identify the point at which the chain of custody was compromised?
2. Why is leaving evidence unattended considered a violation of forensic best practices?
3. How could the defense challenge the integrity of the evidence in court?
4. What standard operating procedure should have been followed to ensure evidence?

Answers:

1A) 2026-10-08 15:00 The sealed USB drive was left unattended on the laboratory desk.

2A) → Anyone could have accessed or tampered with evidence.
→ Continuous custody was not maintained.

3A) They may argue the evidence could have been altered while unattended.

→ This reduces the credibility of the evidence.

4A) → Store evidence immediately in a secure evidence locker.

→ Never leave evidence unattended.

→ Allow access only to authorized personnel.

⑤ ⇒ Unrecorded evidence transfer:

→ A digital forensic team is investigating a ransomware attack on a financial institution. A server hard disk was seized as critical evidence. Review the evidence. Review evidence where chain of custody was broken.

⇒ Evidence log:

2026-11-14 8:45 Server Hard disk seized by officer A from the organization's.
2026-11-14 9:00 SHA-256 hash computed & recorded by officer A: 7e3a..
2026-11-14 9:15 Evidence sealed and stored in the digital evidence locker.
2026-11-15 10:00 Evidence removed from the locker for forensic examination.
2026-11-15 13:35 Forensic Analyst B received the hard disk for imaging.
2026-11-15 13:35 SHA-256 hash verified 7e3a.. (match)
2026-11-15 17:00 Forensic image completed, original hard disk resealed.

⇒ Questions:

1. Identify the point where the chain of custody was broken.
2. Why is the unrecorded transfer of evidence a serious procedural violation?
3. Even though the value matches, why can the defense still challenge the evidence court.
4. What information should have been documented during every evidence transfer to maintain an unbroken chain of custody?

⇒ Answers:

- 1A) 2026-11-15 10:00 Evidence was removed from storage without recording who removed it or why.
- 2A) → The evidence movement cannot be traced.
→ Accountability is lost.
- 3A) → A matching hash proves integrity, but not proper custody.
→ missing documentation creates doubt. handling procedures.
- 4A) Name of person transferring and receiving.
→ Signature of both parties.
→ date and time, Reason for transfer.
→ Evidence condition, Authorization details.